Engineering Hubbard models with gated two-dimensional moiré systems

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Lattice models are powerful tools for studying strongly correlated quantum many-body systems, but their general lack of exact solutions motivates efforts to simulate them in tunable platforms. Recently, a promising new candidate has emerged for such platforms from two-dimensional materials. A subset of moiré systems can be effectively described as a two-dimensional electron gas (2D EG) subject to a moiré potential, with electron-electron interactions screened by nearby metallic gates. In this paper, we investigate the realization of lattice models in such systems. We show that, by controlling the gate separation, a 2D EG in a square moiré potential can be systematically tuned into a system whose ground state exhibits orders analogous to those of the square lattice Hubbard model, including the stripe phase. Furthermore, we study how variations in gate separation and moiré potential depth affect the ground-state orders. A number of antiferromagnetic phases, as well as a ferromagnetic phase and a paramagnetic phase, are identified. We then apply our quantitative downfolding approach to triangular moiré systems closer to current experimental conditions, compare them with the square lattice parameters studied, and outline routes for experimental realization of the phases.

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I. INTRODUCTION

Lattice models provide a fundamental framework for understanding the physics of strongly correlated quantum many-body systems. However, exact solutions to these models are rarely available, motivating experimental efforts to construct quantum simulators that can directly emulate lattice Hamiltonians. A prime example has been with neutral atoms in optical lattices [1]. Traditionally, model engineering has been challenging in condensed matter. However, recent advances in two-dimensional materials have presented exciting new opportunities. A promising platform for such simulations involves stacking and twisting multiple layers of two-dimensional (2D) materials. When confined in two dimensions, electron kinetic energy is suppressed and the electron-electron interaction is enhanced owing to the reduced screening effects from other electrons. In the presence of a moiré potential, the electron kinetic energy is further suppressed. The relative interaction strength can be flexibly tuned by adjusting the twist angle [2] and the surrounding dielectric environment [3,4]. Often metallic gates are present in these systems to tune carrier density, and the gate separation offers additional control over both the range and magnitude of interactions [5,6].

Realizing lattice models with the 2D materials platform has received much recent attention. Various lattice models, such as the triangular [2], honeycomb, and kagome lattices [7], have been theoretically proposed to be realizable using heterobilayers or homobilayers of transition metal dichalcogenides (TMDs). More recently, the simulation of the square lattice models has been proposed in twisted bilayer FeSe [8], GeX/SnX(X = S, Se) [9], carbon allotrope C_{568} [10], and more general homobilayers [11]. Experimental realization of the triangular lattice model has been achieved by stacking a monolayer of WSe_2 on top of a monolayer of WS_2 [12–14].

Quantitative guidance for the realization of these models has been rare, however. In large part this shortage is not surprising. The systems involved tend to be strongly correlated, with long-range Coulomb interactions which are screened by metallic gates. They place high demands on accurate theoretical and computational treatment. Deriving effective model Hamiltonians for such systems requires proper downfolding. Since the engineered system will necessarily have differences from the targeted, idealize model, it is important to have a gauge on whether this difference can alter the desired property in any fundamental way. Reliable quantitative predictions of the properties of the cleanest models are already challenging, and treating the engineered system to discern the difference can be even more difficult.

In this paper, we focus on the engineering of Hubbard models with a 2D EG that lives in a moiré potential. The Hubbard model, originally introduced to study magnetism in materials with narrow 3d bands [15], has been extensively explored due to its simple Hamiltonian, rich phase diagram, and relevance to high-T_c superconductivity [16]. Realizing the square Hubbard model in moiré materials would provide a valuable experimental platform to complement existing theoretical studies, for example by enabling transport measurements to probe superconductivity. In Fig. 1 we provide a schematic overview of the system we study and the basic idea of our work.

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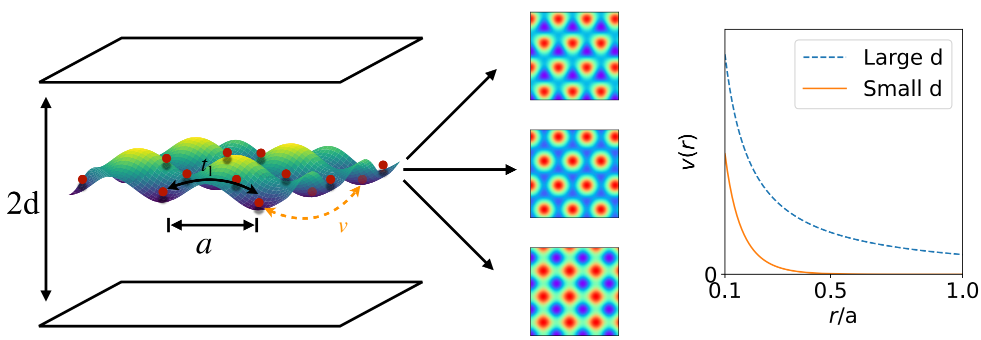

FIG. 1. The schematic of a dual-gate screened 2D EG in a moiré potential. In the left panel, electrons denoted by the red dots are confined in a 2D plane with a moiré potential. The squares above and below the 2D EG represent metallic gates, which screen the electron-electron interaction. The middle three panels are moiré potential of various geometries, where the blue color represents potential valleys and the red color represents potential peaks. From top to bottom, the potential minima form triangular, honeycomb, and square lattices, respectively. The right panel illustrates that the dual-gate screening shortens the range of the electron-electron interaction.

We study the ground-state order of the 2D EG system as a function of the moiré potential depth and the gate separation. We employ a density-functional specifically parametrized for this system [5], and perform density-functional theory (DFT) calculations under the local-density approximation (LDA), more specifically local spin-density approximation (LSDA). We scan the interaction range, strength, and density/filling systematically to identify ground-state orders. With a square moiré potential and gate separation much smaller than the moiré lattice constant, we find ground-state orders that closely resemble those found in the square lattice Hubbard model, including the stripe phase [17,18]. Away from this Hubbard regime, the system exhibits alternative ground-state orders. A phase diagram for $1/8$ hole doping from half filling is shown in Fig. 2, and discussed in further detail in Sec. IV A 2.

To gain intuitive insight into the underlying lattice model and facilitate direct comparison with experiments, we also perform systematic downfolding using Wannier functions to map the 2D EG in both triangular and square moiré potentials onto effective lattice models. We find that in the parameter regime where the stripe phase emerges, both longer-range hopping and interorbital interactions are negligible. The ratio of intraorbital interaction U_0 to nearest-neighbor hopping t_1 in this regime is consistent with those associated with the stripe phase in the square Hubbard model. Notably, the critical U_0/t_1 required to stabilize the stripe phase in the square moiré system is significantly smaller than that in triangular moiré systems realized in current experiments.

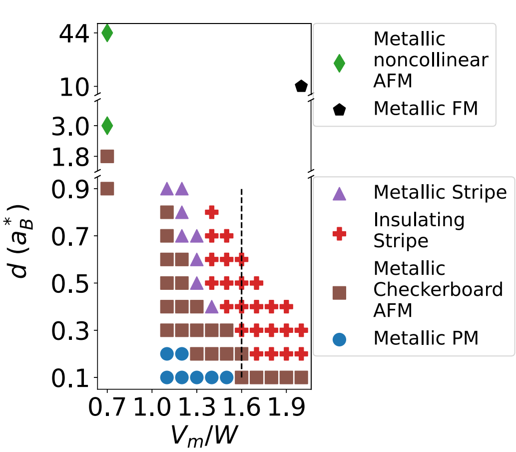

FIG. 2. Phase diagram for the square moiré potential at $1/8$ hole doping and $r_s = 5\, a_B^*$, as a function of the gate distance d and potential depth V_m, normalized by the kinetic energy scale $W = 1/r_s^2$. In the legend, AFM, FM, and PM stand for antiferromagnetic, ferromagnetic, and paramagnetic, respectively. The vertical dashed black line indicates parameter sets we downfold to lattice models for comparison in Sec. IV.

The remainder of this paper is organized as follows. In Sec. II, we describe our system in detail. Section III outlines the computational methods employed in this work. In Sec. IV, we present our main results. We conclude and share some outlooks in Sec. V.

II. DUAL-GATE SCREENED 2D EG IN MOIRÉ POTENTIAL

We consider a 2D EG that lives in a moiré potential, as shown schematically in Fig. 1. Besides intrinsic interest, such a system can model the low energy physics of heterobilayer TMDs, where the valley degrees of freedom is mapped to the electron spins in a 2D EG, in the presence of spin-valley locking [2,19,20]. We use the effective Bohr radius $a_B^* = a_B \times (\epsilon/\epsilon_0)/(m^*/m_e)$ and effective Hartree $E_h^* = E_h \times (m^*/m_e)/(\epsilon/\epsilon_0)^2$ as length and energy units [21], respectively, where m^* is the effective mass of the charge carriers and ϵ the relative permittivity (dielectric constant) of the dielectric spacers. The Hamiltonian of the system is

$$\hat{H} = -\frac{1}{2}\sum_{i=1}^{N} \nabla_i^2 + \frac{1}{2}\sum_{i\neq j} v(|\boldsymbol{r}_i - \boldsymbol{r}_j|, d) - \tilde{V}_m \sum_{i=1}^{N} \Lambda(\boldsymbol{r}_i) + \text{b.g.}, \tag{1}$$

where $v(|\mathbf{r}_i - \mathbf{r}_j|, d)$ is the electron-electron interaction screened by dual metallic gates, separated by a distance d. The moiré potential is characterized by its effective depth $\tilde{V}_m = V_m \frac{m^*/m_e}{(\epsilon/\epsilon_0)^2}$ and shape $\Lambda(\boldsymbol{r})$, which we describe in further detail below. The negative sign in front of V_m maps a system of electrons to a system of holes without having to invert the kinetic energy term. The final term in Eq. (1) accounts for a uniform background charge that ensures overall charge neutrality.

The presence of metallic gates enables tunability of both the range and strength of the Coulomb interaction via the gate separation, as illustrated in the right panel of Fig. 1. In momentum space, the dual-gate screened Coulomb interaction admits a closed-form expression [5]

$$v(k, d) = \frac{2\pi \tanh(kd)}{k}, \tag{2}$$

where k is the magnitude of the wave vector and d is the gate separation. The $\tanh(kd)$ factor suppresses long-range interactions, effectively truncating the Coulomb tail when d is small, allowing the interaction to be tuned from long ranged to short ranged.

The presence of the moiré potential can effectively enhance the electron-electron interaction. As the moiré lattice constant and potential depth increase (at fixed electron density), electrons become more localized at the potential minima, thereby amplifying interaction effects. The shape of the moiré potential can be written as a Fourier expansion on the reciprocal space lattice of the moiré supercell. The lowest order (harmonic) approximation reads

$$\Lambda(\boldsymbol{r}) = \sum_{j=1}^{N/2} 2\cos(\boldsymbol{r}\cdot\boldsymbol{g}_j + \phi), \tag{3}$$

where $\mathbf{g}_j$ are the N smallest reciprocal lattice vectors of the moiré superlattice. In the triangular moiré potential, the phase parameter ϕ changes the number of local minima of moiré potential from one to two and back every 120°. For the square moiré potential, ϕ only results in a translational shift of the potential, and we set $\phi = 0$. Examples of moiré potentials with different geometries are shown in the middle panel of Fig. 1.

The interaction strength also depends on the electron density. To make this dependence explicit, we can use the Wigner-Seitz radius r_s of a homogeneous 2D EG as length units, defining $\tilde{\boldsymbol{r}} = \boldsymbol{r}/r_s$. For simplicity, we express r_s in units of the effective a_B^*. In r_s units, the Hamiltonian reads

$$\begin{aligned}\hat{H} = &-\frac{1}{2r_s^2}\sum_i \tilde{\nabla}_i^2 + \frac{1}{2r_s}\sum_{i\neq j} v(|\tilde{\mathbf{r}}_i - \tilde{\mathbf{r}}_j|, d)\\ &- \tilde{V}_m \sum_i \Lambda(\tilde{\mathbf{r}}_i) + \text{b.g.},\end{aligned} \tag{4}$$

which explicitly shows the scaling of the kinetic and interaction terms with r_s. We use the prefactor of the kinetic term as a proxy to the band width: $W = 1/r_s^2$. The electron-electron interaction dominates at low densities (large r_s).

III. METHOD

Numerical methods play a vital role in complementing ongoing experimental efforts. Direct detection of magnetic texture remains an experimental challenge. Numerical simulations can readily access spatially resolved magnetic properties, making them especially valuable for exploring and identifying interesting regions of parameter space for magnetic applications.

We apply two separate approaches which complement each other. In the first, we perform LSDA calculations on the moiré continuum Hamiltonian of Eq. (4). The resulting ground-state properties are presented as a function of the parameters and compared with known properties in the Hubbard model. In the second approach, we apply Wannierization and downfolding to the continuum Hamiltonian to obtain a lattice model, and examine the resulting model parameters. The results from the two approaches provide cross-checks, which also lead to additional insights on the behavior of the gated 2D EG in the context of model engineering. In this section, we introduce our two computational approaches, describing the DFT with screened interactions in Sec. III A, and the Wannierization and downfolding on top of the DFT results in Sec. III B.

A. Density functional theory with screened interactions

To study the gated electron gas in a moiré potential, we apply DFT with screened interactions. Practical DFT calculations rely on approximate forms such as LSDA. The conventional form of the exchange-correlation functional $E_{xc}[n, p]$ is parametrized in terms of the local electron density n and spin polarization p via fits to accurate total energies obtained from quantum Monte Carlo (QMC) calculations [22–24]. However, this form does not account for metallic gate screening, which is essential for modeling gated moiré systems with tunable short-range interactions. To overcome this limitation, we adopt the recently developed exchange-correlation functional $E_{xc}[n, p, d]$ parametrized from QMC calculations of dual-gate screened homogeneous 2D EG, which incorporates the gate separation d as an additional variable [5]. This functional was implemented within the framework of the libxc library [25]. All DFT calculations are carried out using a modified version of Quantum Espresso adapted for 2D systems [26,27].

To determine the ground state, we use simulation cells with various geometries and initialize each self-consistent field (SCF) run with approximately 100 randomly generated initial configurations. In each configuration, electrons are placed in localized s orbitals centered at random positions. For collinear spin calculations, each electron is assigned a spin orientation (up or down) with equal probability. In addition to collinear spin calculations, we also perform noncollinear spin DFT simulations using the generalized LSDA formalism introduced in Ref. [28]. Most of the noncollinear calculations were initialized with the spin polarization vectors randomly oriented within the 2D plane, but we have checked robustness in select cases with other initial states. After all SCF runs, the ordered state with the lowest total energy among the converged solutions is selected as the ground state. The kinetic energy cutoff is chosen based on density ($270/r_s$ Ry). To ensure consistent Brillouin zone sampling across different system sizes, we fix the product of the number of k points and the number of unit cells along each lattice direction to 32. Additionally, a [0.5, 0.5] shift is applied to the Γ-centered Monkhorst-Pack grid to improve sampling. We also performed calculations constrained to the paramagnetic phase using a 4×4 supercell for Wannierization and downfolding (see Sec. III B).

The plane-wave kinetic energy cutoff is chosen based on density ($100/r_s$ Ry), and the Brillouin zone is sampled using a 6×6 Γ-centered Monkhorst-Pack grid. The convergence threshold for total energy is set to 0.05 mHa in all DFT calculations.

In previous studies, the Hartree—Fock (HF) method has been widely applied to moiré systems [29,30]. Compared to HF, DFT with LSDA offers significantly greater computational efficiency, particularly for large simulation cells. Furthermore, DFT/LDA often yields more accurate results in practice, as HF tends to overemphasize symmetry-broken, ordered states; an example in 2D EG moiré systems is given recently [19]. We expect the accuracy of DFT to sustain, possibly even improve, in moiré systems as the gate screening becomes stronger, which reduces the influence of long-range interactions and thereby makes the system more amenable to local approximations such as LDA.

B. Wannierization and downfolding

To gain insights and facilitate comparison with effective lattice models, we employ two standard postprocessing steps: Wannierization and downfolding. Wannierization transforms the extended Bloch states into localized real-space orbitals, while downfolding systematically reduces the full Hilbert space to a low-energy subspace that retains the essential physics of interest. Together, these procedures enable a more transparent interpretation of the electronic structure in terms of effective lattice degrees of freedom.

During the Wannierization process, the Kohn—Sham orbitals, initially defined in momentum space and labeled by Bloch wave vectors, are transformed into real-space localized orbitals. The spatial localization or spread of the resulting Wannier functions can be tuned by applying a unitary transformation to the set of Kohn—Sham orbitals at each **k**-point. This unitary transformation is determined by minimizing the total spread of the Wannier functions, following the procedure introduced in Ref. [31] and implemented in the Wannier90 code [32]. The resulting maximally localized Wannier orbitals serve as a natural basis for constructing effective lattice models.

In downfolding [33,34], we map the 2D EG in a moiré potential to a lattice model. In the DFT band structure for the paramagnetic phase, we identify a set of isolated energy bands that are well separated from higher-energy bands. The number of these isolated bands equals the number of moiré unit cells in the calculation, N. These N Kohn–Sham orbitals are Wannierized following the procedure described above. In this way, we obtain a lattice model in the subspace of the original Hilbert space, where the number of lattice sites equals the number of moiré unit cells in the original model.

The one-body elements in the resulting low-energy Hamiltonian are obtained from the associated DFT-based tight-binding model in the local Wannier basis. To incorporate dynamic screening effects, we employ the constrained random phase approximation (cRPA). In this approach, the bare Coulomb interaction within the low-energy (active) subspace is retained, while screening contributions from the remaining (virtual) subspace are computed explicitly at the RPA level. We perform DFT calculations that include a large number of unoccupied bands to evaluate the RPA frequency-dependent dielectric function, and extract the effective interaction parameters by taking the zero-frequency limit [33,35,36]. The cRPA downfolding approach improves the resulting Hamiltonian over that from the bare matrix elements, and results in a downfolded lattice model that retains the essential low-energy physics of the original 2D electron system in a moiré potential.

IV. RESULTS

In this section, we present our results, organized into three parts. We study the ground state orders of the system described in Sec. II, beginning with the square lattice geometry at small gate separation. We then explore the effects of increasing gate separation, followed by a reduction in moiré potential depth, and finally consider the triangular geometry to establish connections with more commonly studied experimental setups. We compare the results of our continuum Hamiltonian calculations with results from the Hubbard model. In addition, we perform systematic downfolding and Wannierization to obtain model parameters from the original continuum system, to help validate and gauge the connection with the Hubbard model.

In Sec. IV A below, we show that standard square lattice Hubbard model with near-neighbor hopping and on-site interaction can be realized at very small d. Note that the statement that d is—small—is meant in a relative sense, since our length unit is the effective Bohr radius defined in Sec. II, where m^* and ϵ are both set to unity for computational convenience. The actual values of these parameters depend on the specific material realization in an experimental setup. We examine both half filling and 1/8-doping using DFT/LSDA with our parametrized functional, and show that the ground state of this system reproduces key features of the corresponding ground state in the Hubbard model, such as antiferromagnetism and stripe phases. Through the phase diagram of the continuum model and the downfolding analysis, we examine deviations from the simple Hubbard model as d increases and the electron interaction becomes longer ranged, as we investigate in Sec. IV B. Our calculations reveal metallic ground states, with ferromagnetism in a deep moiré potential and noncollinear antiferromagnetic (AFM) order in a shallow moiré potential. We again obtain via downfolding the model parameters, which correspond to the extended Hubbard models. Finally, in Sec. IV C we apply the same methodologies to the triangular geometry. We establish a quantitative mapping from current experiments on triangular moiré potentials to the corresponding lattice models. We also summarize, in Table I, our downfolding results for some of the moiré systems that have been proposed for realizing the Hubbard model.

A. Realization of the Hubbard model

We study the 2D EG in a square moiré potential at both half filling ($\nu = 1$) and 1/8 hole doping ($\nu = 7/8$). In the conventional Hubbard model, half filling corresponds to one electron per lattice site, with each site hosting a single orbital that can accommodate up to two electrons with opposite spins. By contrast, in our moiré system, each moiré potential valley contains multiple energy levels. We use $\nu = 1$ (7/8) to

TABLE I. Downfolding results for some of the candidate moiré systems for Hubbard model realization. For the $MoSe_2/MoS_2$ system, m^* is taken as the average along the $K-\Gamma$ and $K-M$ directions for the $C7$ stacking configuration studied in Ref. [37]. For $MoSe_2/WS_2$, m^* is taken as the average along the x and y directions in the AC stacking in Ref. [38]. The V_m and ϕ for both systems are taken from the AA stacking in Ref. [39]. In the U_1/U_0 column, ∼0 means smaller than our numerical resolution.

Systems	θ (°)	a_M (nm)	V_m (meV)	ϕ (°)	m^*	ϵ	d (nm)	U_0/t_1	U_1/U_0	$t_{\sqrt{3}}/t_1$	U_0 (meV)
WSe_2/MoS_2 [2]	0	8.5	5.1	−71	0.35	4.2	20.0	26.4	0.50	0.23	55.7
WSe_2/MoS_2 [2]	0	8.5	5.1	−71	0.35	4.2	5.0	18.1	0.35	0.23	38.1
WSe_2/MoS_2 [2]	0	8.5	5.1	−71	0.35	4.2	2.0	11.4	0.24	0.23	24.7
WSe_2/MoS_2 [2]	0	8.5	5.1	−71	0.35	4.2	1.1	8.0	0.18	0.24	16.9
WSe_2/MoS_2 [2]	0	8.5	5.1	−71	0.35	4.2	0.8	6.4	0.15	0.24	14.2
WSe_2/MoS_2 [2]	0	8.5	5.1	−71	0.35	3.0	0.8	7.9	0.18	0.24	17.5
WSe_2/MoS_2 [39]	0	8.5	11	40	0.35	4.2	0.8	15.2	0.06	0.17	25.2
WSe_2/MoS_2 [39]	0	8.5	11	40	0.35	3.0	0.8	18.7	∼0	0.17	31.3
$MoSe_2/MoS_2$ [37,39]	0	8.3	9	42	0.38	4.2	0.8	12.3	∼0	0.19	21.9
$MoSe_2/MoS_2$ [37,39]	0	8.3	9	42	0.38	3.0	0.8	15.1	∼0	0.19	27.0
$MoSe_2/WS_2$ [38,39]	0	8.2	7	35	0.64	4.2	0.8	28.4	∼0	0.15	25.4
$MoSe_2/WS_2$ [38,39]	0	8.2	7	35	0.64	3.0	0.8	33.7	∼0	0.15	30.1
$WSe_2/MoSe_2$ [2]	1	19	6.6	−94	0.35	4.2	0.8	352	∼0	0.01	16.0

represent the case where each moiré unit cell is occupied by 1 (7/8) electrons on average.

At half filling, the Hubbard model is known to exhibit a Mott insulating ground state with checkerboard AFM for any finite interaction strength U/t. Away from half filling, however, the model exhibits increased complexity and lacks an exact solution. Recent numerical studies [40] indicate that the stripe phase is the ground state at 1/8 hole doping, when U/t exceeds a critical threshold [17].

Below we discuss the two fillings in two subsections. We observe in the 2D EG system both the checkerboard AFM phase and the stripe phase. The properties of these phases are consistent with state-of-the-art results from the Hubbard model. Moreover, the parameter values obtained through downfolding are consistent with a simple Hubbard model, with hopping amplitudes t_n for $n > 1$ (beyond near neighbor) and interaction strength U_n for $n > 0$ (beyond on site) both strongly suppressed. In the stripe phase, the critical ratio U_0/t_1 obtained through downfolding agrees quantitatively with that of the square Hubbard model, reinforcing the validity of our moiré-engineered realization.

1. Half filling ($\nu = 1$)

For $\nu = 1$, we systematically explore the parameter space defined by $0.1 \leqslant d \leqslant 0.9$ (in units of a_B^*) and $1.1 \leqslant V_m/W \leqslant 2.0$. We find that the insulating AFM phase is robust in most of this region. An example of the corresponding charge and spin configuration is shown in Fig. 3. This result is consistent with the half filled Hubbard model, whose ground state is known to be a Mott insulator with AFM order for any finite U/t.

However, a notable difference arises in the moiré system: within a narrow window at small gate separation ($d = 0.1\ a_B^*$) and moiré potential strength ($V_m/W = 1.1$–1.3), we observe a paramagnetic metallic phase. Our DFT/LSDA results indicate that the transition from the AFM to the paramagnetic phase coincides with an insulator-to-metal transition.

We quantitatively connect the continuum moiré model to an effective lattice model through the downfolding procedure described in Sec. III B. The downfolded parameters reveal that the critical effective interaction strength U_0/t_1 for the AFM insulator to the paramagnetic metal phase transition is between 2.2 and 2.5. Notably, we also find that the diagonal (next-nearest-neighbor) hopping $t_{\sqrt{2}}$ is significantly suppressed compared to the third-nearest-neighbor hopping t_2. For instance, at $V_m/W = 1.6$ and $d = 0.3\ a_B^*$, $t_{\sqrt{2}}/t \approx 0.003$ while $t_2/t \approx 0.08$; at $V_m/W = 1.1$ and $d = 0.1\ a_B^*$, $t_{\sqrt{2}}/t \approx 0.001$ while $t_2/t \approx 0.11$. This suppression arises from the presence of potential maxima located along the diagonals between neighboring potential minima, which effectively inhibit electron tunneling along these paths.

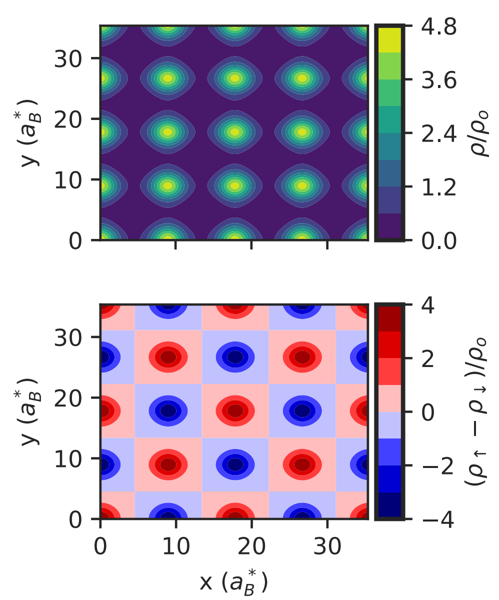

FIG. 3. Charge and spin patterns in the ground state of system $\nu = 1$, $r_s = 5\ a_B^*$, $V_m/W = 1.6$, and $d = 0.3\ a_B^*$. The upper panel is the charge density and the lower panel the spin density. Both densities are normalized by the average charge density $\rho_0 = N_e/A$, where N_e is the number of electrons and A the area of the simulation cell.

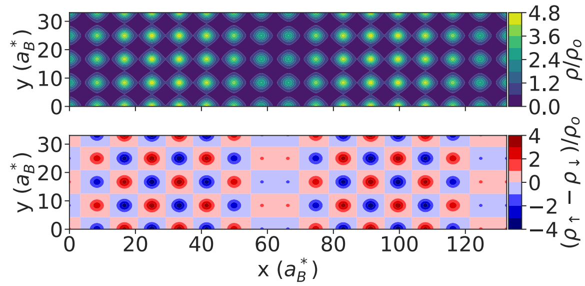

FIG. 4. Charge and spin patterns in the ground state of system $\nu = 7/8$, $r_s = 5\ a_B^*$, $V_m/W = 1.6$, and $d = 0.3\ a_B^*$. Setting and conventions are similar to Fig. 3.

2. 1/8-*doping* ($\nu = 7/8$)

We study our square moiré system at filling $\nu = 7/8$, i.e., 1/8-hole doped in the convention of the Hubbard model. The phase diagram scanning a range of gate separation d and moiré potential depth is shown in Fig. 2. In this section, we focus on small d and deep potential, in particular $V_m/W = 1.6$. We find a stripe phase at $d = 0.3\,a_B^*$ as illustrated in Fig. 4. In the charge density plot, we identify periodically spaced regions of suppressed charge density. Simultaneously, the spin density shows checkerboard AFM order, interrupted by domain walls that lie within the stripe regions. These domain walls introduce a π phase shift in the AFM order. This state is consistent with the stripe phase found in the Hubbard model at zero temperature [17,40]. Following the literature, we refer to this as a linear stripe phase [18,40]. Within each stripe period along a row, there is approximately one hole on average, characterizing this configuration as a filled stripe [40].

As the gate distance d increases, the system undergoes a transition to a diagonal stripe phase, where both charge and spin modulate along the [1, 1] direction (see Fig. 5).The modulation patterns are similar to the linear stripe case, but the stripe orientation rotates by 45°. Additionally, the charge density minimum shifts from bond centered (in the linear stripe) to site centered (in the diagonal stripe). This is also manifested in the positions of the domain walls in the spin density plot. However, since the stripe pattern is spatially extended over several sites, the distinction between bond-node and site-node stripes becomes less critical [40].

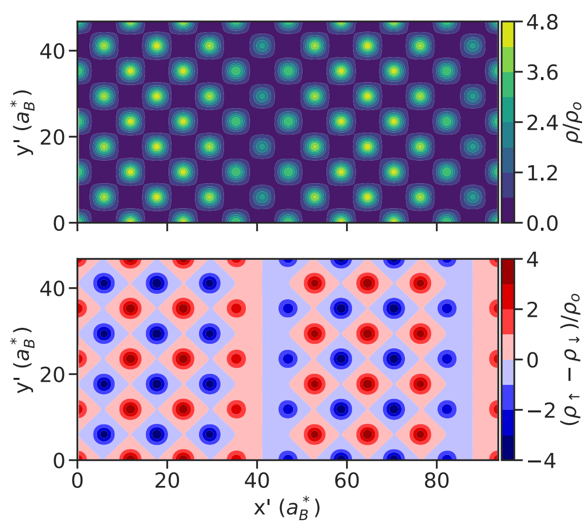

FIG. 5. Diagonal charge and spin patterns in the ground state of system $V_m/W = 1.6$ and $d = 0.4\ a_B^*$. In this plot, the horizontal and vertical axes, denoted by x' and y', represent the (1,1) and $(-1, 1)$ directions, respectively. That is, they are rotated by 45° with respect to the x and y axes in Figs. 3 and 4; other conventions are the same.

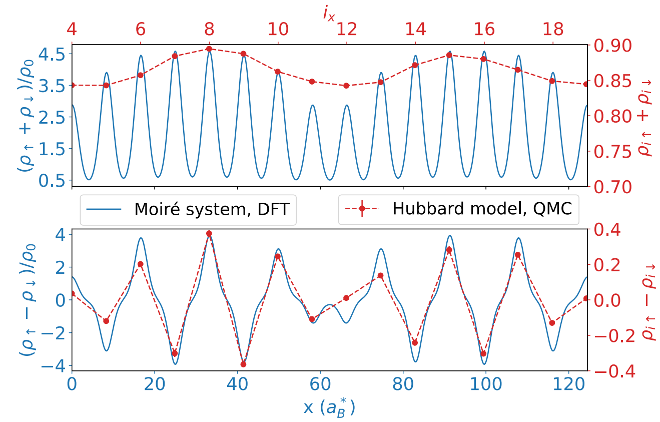

FIG. 6. Charge and spin density line cuts in the ground state of system $\nu = 7/8$, $r_s = 5\ a_B^*$, $V_m/W = 1.6$, $d = 0.4\ a_B^*$. The upper panel is the charge density and the lower panel the spin density. The blue curves are DFT results for the moiré system. For comparison, the red curves are results from CP AFQMC [41,42] calculations in the square-lattice Hubbard model on a (L_x, L_y)=(4, 32) at $U/t = 8$ (i_x and i_y label the lattice sites along the x and y directions.) Error bars are smaller than the marker size. In both panels, the Hubbard model sites are aligned with the minima of the moiré potential. The slight asymmetry of the red curves arises from the effect of the pinning fields in the finite system.

When mapped to a lattice model through Wannierization and downfolding, we find that these parameter values of d and V_m/W correspond to U/t in the range of 4 to 8, with essentially no interaction beyond on site and only small hopping amplitude t_n/t_1 beyond near neighbor (further details in the next section). The ground-state orders seen above are compatible with the stripe phase found in the Hubbard model at zero temperature [17,40]. An example of a quantitative comparison is given below. By scanning d, we find the range of the critical value U/t for the onset of the stripe order in agreement with the phase diagram of the Hubbard model (no next-nearest-neighbor hopping) which places the stripe phase boundary at 1/8-doping between $U/t = 3.5$ and 5 [17]. Moreover, we scan the range $1.1 \leqslant V_m/W \leqslant 2.0$ and find that such an agreement holds for $V_m/W \geqslant 1.6$. The disagreement for $V_m/W < 1.6$ is likely due to the increasing influence of further neighbor hoppings.

As an example, we show in Fig. 6 a comparison between the DFT result in our moiré system and results in the Hubbard model from constrained-path auxiliary field quantum Monte Carlo (CP AFQMC) [41,42] calculations. The CP AFQMC results are obtained following the same procedures as in Refs. [17,40,43]. Open (periodic) boundary condition is applied in y (x) direction, and pinning fields ($v_p = 0.1$) are applied on the two edges of the cylinder. The trial wave function is the unrestricted HF solution to the same model with an effective on-site interaction $U_{\mathrm{eff}} = 2.7$, determined from a self-consistent approach [43]. We find that the envelopes of the charge and spin density in the moiré system closely reproduce those of the Hubbard model when the moiré potential minima are mapped onto the Hubbard sites. The continuum model offers additional details. Both the charge and spin densities are strongly localized at the potential valleys. Between neighboring valleys, the spin density decreases to nearly zero, whereas the charge density remains finite at approximately 0.5.

A natural question can arise regarding the reliability of DFT/LSDA, in treating the Hamiltonian in Eq. (4). There are two aspects which provide strong support for the validity of our conclusions. First, since the system is a 2D EG in an external potential which is rather smooth, it is reasonable that an LSDA treatment based on a functional specifically parametrized for the homogeneous 2D EG (with the exact same gate-screened interaction) can capture the charge and spin properties. Indeed a direct comparison under the triangular geometry between DFT/LSDA and QMC calculations shows that the former is rather accurate in predicting the ground-state phases [19]. Second, in the Hubbard model it is known that unrestricted Hartree-Fock gives reasonable predictions on the stripe order [17,18], especially with some renormalization of the interaction strength U/t [17,43]. In some sense, our LSDA calculation can be compared with the unrestricted HF (UHF) in the Hubbard model. Interestingly, the diagonal stripe order we observe above in Fig. 5 is also found in the Hubbard model at larger U/t in UHF [18], while its existence is more ambiguous in more accurate QMC calculations [17]. These observations all indicate that the system indeed realizes the standard square lattice Hubbard model.

B. Larger d and the extended Hubbard model

The results from the previous section confirm the realization of the square lattice Hubbard model at small d. Here we stay at $\nu = 7/8$ and scan a larger range of d as well as the moiré potential depth. The phase diagram is shown in Fig. 2. The two systems discussed in Sec. IV A 2 fall on the vertical dashed line, where stripe phases are seen, with charge and spin densities exhibiting modulations along the [1, 0] direction at $d = 0.3$ and along [1, 1] at $d = 0.4\,a_B^*$. At even larger d, we observe a transition to a noncollinear stripe phase. For example, at $V_m/W = 1.4$, $d = 0.8\,a_B^*$ and $V_m/W = 1.5$, $d = 0.7\,a_B^*$, the ground state exhibits noncollinear spin textures superimposed on a stripelike charge distribution, as shown in Fig. 7. These states are insulating from our DFT/LSDA band structure.

We also observe that the stripe phase becomes metallic when the moiré potential becomes shallower. These metallic stripe phases exhibit longer periodicities compared to their insulating counterparts and are marked by purple triangles in Fig. 2. As the gate distance d decreases, the stripe phase transitions into a metallic checkerboard AFM phase. A paramagnetic phase emerges as the interaction strength is further reduced. At $d = 0.1\,a_B^*$, decreasing the moiré potential from $V_m/W = 1.6$ to 1.5 causes the ground state to transition from a metallic checkerboard AFM phase to a metallic paramagnetic phase. This is in contrast to the half filling case studied in Sec. IV, where the insulator-to-metal transition is accompanied by a transition from the AFM to the paramagnetic phase.

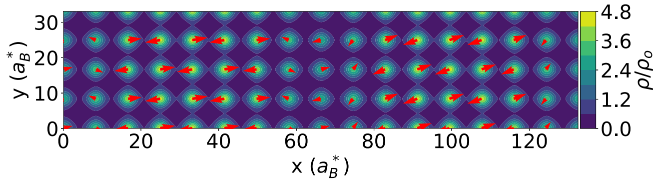

FIG. 7. Charge and spin patterns in the ground state of system $\nu = 7/8$, $r_s = 5\,a_B^*$, $V_m/W = 1.5$, $d = 0.7\,a_B^*$. The red arrows denote the spin orientations in the x-y plane. The maximum out-of-plane (z) spin component is comparable to the maximum y spin component.

We map the main portion of the parameter space in Fig. 2 into an effective lattice model through Wannierization and downfolding. We follow the same procedure as described in Sec. III B and applied in the $\nu = 1$ case above. The downfolded parameters along the vertical line cut in Fig. 2 are presented in Fig. 8. The results show that hopping amplitudes beyond nearest neighbors are small, and interorbital interactions are negligible, as mentioned in Sec. IV A 2. The downfolded Hamiltonian corresponds to a t-t'-t''-U Hubbard model, with very small t' ($t'/t \sim 0.002$) and a $t''/t \sim 0.09$. Such a small but non-negligible t'' might help stabilize the metallic checkerboard phase for small U_0/t_1. To have a more global view of the model parameters in the space of $0.1 \leqslant d \leqslant 0.9$ and $1.1 \leqslant V_m/W \leqslant 2.0$, we plot the corresponding values of the interaction ratio U_0/t_1, the nearest-neighbor interaction ratio U_1/U_0, and the next-nearest-neighbor hopping ratio t_2/t_1 as heat maps in Fig. 9. We see that throughout this parameter region, the ratio of interorbital interactions to t_1 and the hopping amplitudes beyond the nearest neighbors are negligible compared to U_0/t_1 and t_1, respectively, indicating that our moiré system still approximates a Hubbard model reasonably well.

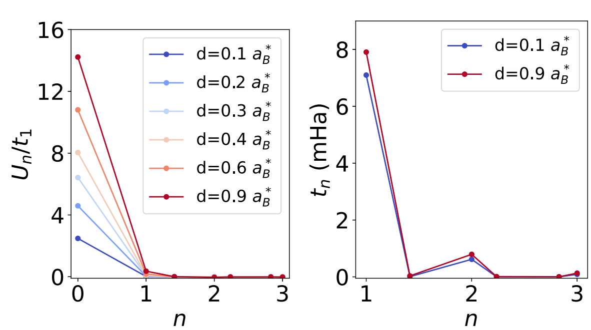

FIG. 8. The extended Hubbard model parameters obtained from downfolding at $\nu = 7/8$, $r_s = 5\,a_B^*$, and $V_m/W = 1.6$. The calculations are performed on a 4×4 cell, and follow the procedure described in Sec. III B and applied in the $\nu = 1$ case. The system parameters are denoted by the vertical dashed black line in Fig. 2. The left panel displays the orbital interactions U_n normalized by the nearest-neighbor hopping amplitudes t_1. The right panel plots the corresponding hopping amplitudes as a function of the orbital separation in the number of lattice constants n. The moiré potential peaks suppress the diagonal-related hopping.

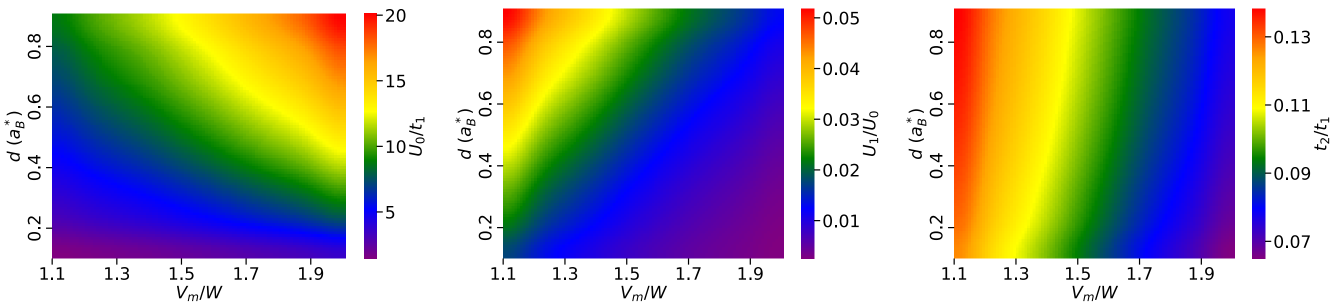

FIG. 9. Heat maps of the parameters from downfolding as functions of gate distance d and the moiré potential depth normalized by the kinetic energy V_m/W: Intraorbital interaction U_0 normalized by the nearest-neighbor hopping amplitude t_1 (left), the nearest-neighbor interaction U_1 normalized by U_0 (center), and the third-nearest-neighbor hopping amplitude t_2 normalized by t_1 (right).

We have also investigated two other regimes with even larger gate separation d: shallow and deep moiré potentials. In the former, a shallow moiré potential ($V_m/W = 0.7$), we observe a transition from a metallic checkerboard AFM phase to a metallic noncollinear AFM phase as d increases from 1.8 a_B^* to 3 a_B^*. The corresponding charge and spin order are shown in Fig. 10. In the noncollinear AFM phase, the charge density retains the discrete translational symmetry of the moiré potential, while the spin order becomes noncollinear with the spin orientations modulating along the [1,1] direction. The downfolding parameters associated with this transition are plotted in Fig. 11. We can see that the hopping amplitudes are not significantly affected by the gate screening, while both the on-site and longer-range interactions increase with d, as expected.

In the case of large d and a deep moiré potential, the ground state becomes a ferromagnetic metal at $V_m/W = 2.0$, $d = 10\, a_B^*$. The downfolding parameters for this case are shown in Fig. 12 together with those for the system with $V_m/W = 0.7$ and $d = 3\, a_B^*$. In the former, U_0/t_1 is approximately 5 times larger. A strong on-site interaction can favor ferromagnetism, as is given by the Stoner criterion [44]. Such a phase has been shown to appear in HF calculations on the Hubbard model [18]. As discussed earlier, in the strong interaction regime, it is plausible that our LSDA calculation in the 2D EG with external potential behaves in a manner more similar to UHF (than to more accurate many-body results).

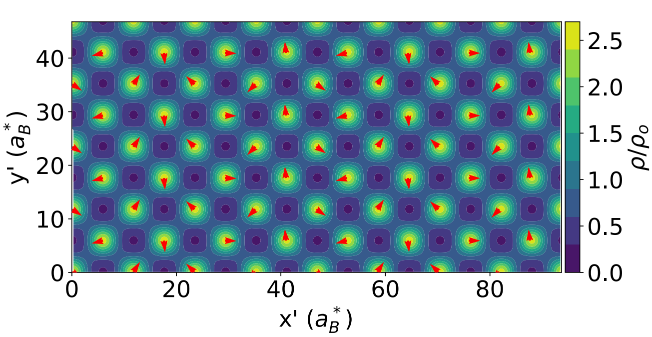

FIG. 10. Charge and spin patterns in the ground state of system $\nu = 7/8$, $r_s = 5\, a_B^*$, $V_m/W = 0.7$, and $d = 3\, a_B^*$. In this plot the horizontal and vertical axes, denoted by x' and y', represent the (1,1) and $(-1, 1)$ directions, respectively. That is, they are rotated by 45° with respect to the x and y axes in Fig. 7. The red arrows denote the spin orientation in the y-z plane. The maximum out-of-plane (x) spin component is around 1/5 of the maximum of the $y - z$ plane component.

C. Triangular lattice

Experiments in moiré systems have already specifically targeted the triangular Hubbard model [12–14]. In these experiments, the holes in the valence band top of WSe_2 play the role of electrons (with the effective mass 0.35) [2]. These holes experience a moiré potential arising from the lattice mismatch between the top layer of WS_2 and the bottom layer of WSe_2, with a moiré lattice constant a_M of about 8 nm. The interaction between the holes are screened by metallic gates. Hexagonal boron nitride (hBN) layers separate the metallic gates from the moiré system, providing a dielectric constant ϵ of either 3 [12,14] or 4.2 [13]. Following the treatment in Eq. (4), we find this results in an effective Bohr radius a_B^* of approximately 8.6 and 12.0, respectively. We summarize three experimental realizations of the triangular Hubbard model in WS_2/WSe_2 bilayers in Fig. 13. The variation in r_s (in units of a_B^*) arises from differences in the dielectric constant ϵ of the dielectric layers.

To quantitatively connect the existing experiments on the triangular moiré potential to lattice models, we again apply DFT and downfolding as we have done in the square geometry above. As a representative example, we consider the experiment of Tang *et al.* in Ref. [14] (referred to as EXP. 1). The bottom gate placed approximately 0.8 nm from the 2D system, while the top gate was kept at a distance of

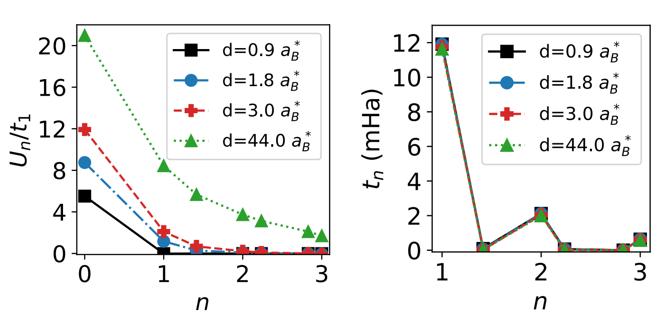

FIG. 11. The downfolding parameters for systems at $\nu = 7/8$, $r_s = 5\, a_B^*$, $V_m/W = 0.7$, and $d = 0.9,\ 1.8,\ 3.0,\ $ and $44.0\, a_B^*$. The layout is similar to Fig. 8.

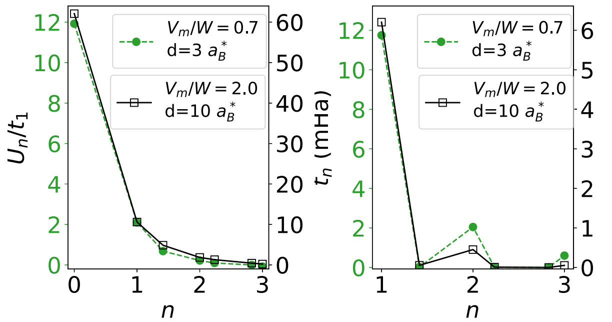

FIG. 12. The downfolding parameters for systems $\nu = 7/8$, $r_s = 5\ a_B^*$, with $V_m/W = 0.7$, $d = 3\ a_B^*$ and $V_m/W = 2.0$, $d = 10\ a_B^*$, respectively. The layout is similar to Fig. 8.

20 nm. To apply our density functional parametrized for a symmetric-dual-gate-screened system, we considered a system with both gates 0.8 nm away from the 2D system. This corresponds to 1.8 a_B^* when we take $\epsilon \approx 3$ for the dielectric constant hBN [12,14]. Accordingly, $r_s = 9.02 a_B^*$ at $\nu = 1$. We use the moiré potential $V_m = 15$ meV and the parameter $\phi = 45^\circ$ in Eq. (3), as fitted by Zhang *et al.* [39] from DFT results assuming relaxed interlayer distance and rigid in-plane lattice. In Hartree atomic units, this corresponds to $V_m = 14.2$ mHa*.

The downfolding results are presented in Fig. 13. Our calculation show that while gate screening does shorten the electron-electron interaction range in EXP. 1, the on-site interaction remains substantial, with $U_0/t_1 = 18$. So this is more consistent with the triangular lattice Hubbard model realized in the strong interaction limit. In contrast, when the gate separation is larger ($d > 20$ nm), the interactions become long ranged and the ratio U_0/t_1 increases by a factor of 3. These situations are also distinct from the parameter regime we were targeting above, and realize extended Hubbard models in the triangular lattice.

To quantify how the interaction and hopping parameters of lattice models realized in moiré systems are controlled by experimentally accessible knobs, we perform downfolding for several theoretically proposed systems. We set d to be equal or above the smallest d (0.8 nm) realized in experiments on WS_2/WSe_2 bilayers (EXP. 1). The results are summarized in Table I. We find that moiré-material—dependent properties, such as the moiré lattice constant, moiré potential, and the effective mass of the charge carriers, jointly determine the interaction strengths together with the gate separation and dielectric environment. The moiré potential in realistic devices can be larger than the theoretical values adopted here from the literature [45]. For a given moiré system and a fixed lower bound for d, the interactions can be efficiently reduced by increasing the ϵ of the dielectric environment. In contrast, the hoppings are insensitive to variations in ϵ and d.

V. CONCLUSIONS AND OUTLOOK

In this paper, we presented a systematic and quantitative study to show how a 2D EG in a moiré potential can serve as a tunable quantum simulator for lattice models, by controlling the potential and the metallic gates. In particular, with a square moiré potential, the system can be made to approach the iconic Hubbard model which has been a hallmark in modern condensed matter physics. We showed that stripe phases appear and show properties consistent with those in the ground state of the square lattice Hubbard model. Downfolding the moiré system to a lattice model, we find that the parameters are consistent with the Hubbard model which exhibits similar phases. As the range and strength of the electron-electron interactions are varied via gate separation and moiré well depth, we observe a rich variety of metallic magnetic orders, including checkerboard AFM, noncollinear AFM order, paramagnetism, and ferromagnetism. We obtain the corresponding lattice models via downfolding. Comparing with the triangular moiré systems already realized in experiments, weaker and shorter range interaction is needed for realizing the square lattice simple Hubbard model for studying the stripe phases and potential superconducting phases [46]. The simple harmonic potential used in this work can be a reasonable approximation for moiré materials whose charge carriers live in one layer, experiencing a modulated potential with long periodicity. Our results highlight the potential of 2D moiré systems as a quantum simulator, which can complement alternative analog quantum computing platforms such as

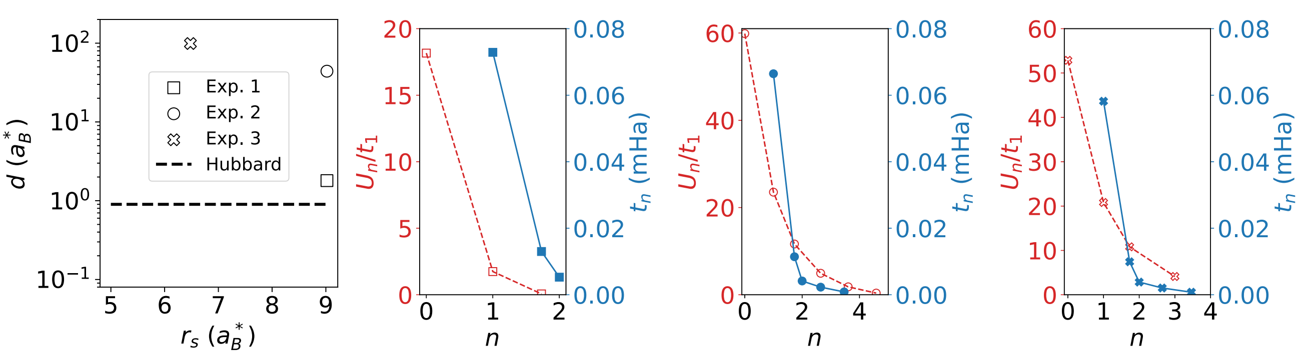

FIG. 13. The downfolding parameters for the existing experiments on triangular moiré potential. Hubbard physics appears for $d < 0.9\ a_B^*$ in Fig. 2. This region is below the horizontal dashed black line in the first panel. Three sets of experimental parameters on triangular moiré potential are represented by the open markers, where Exp. 1 (second panel) and Exp. 2 (third panel) are taken from Refs. [12,14], and Exp. 3 (fourth panel) from Ref. [13].

neutral cold atoms [1]. Compared to the latter, the 2D materials systems offer opportunities for different measurements.

We employed two theoretical approaches in this study. The first is a direct treatment of the moiré continuum Hamiltonian with DFT, using an LSDA functional tailor-made from QMC to account for the presence of metallic gates [5]. The second is *ab initio* downfolding of the continuum Hamiltonian into a lattice model. With the first, we compare the ground-state phases and properties to what is known in the Hubbard model from both UHF [18,47] and more accurate QMC calculations [17,48] to identify potential matching between the continuum system and the model. The second approach then provides model parameters for another validation and cross-check. It will be valuable to further evaluate the accuracy of DFT/LSDA by benchmark against direct calculations in the continuum by more advanced many-body methods, such as QMC [19,20,49]. However, the cross-check with both UHF and QMC results in the Hubbard model, and the consistency between our two approaches provide a high level of confidence and allowed us to make quantitative predictions.

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DATA AVAILABILITY

The data that support the findings of this article are not publicly available. The data are available from the authors upon reasonable request.

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